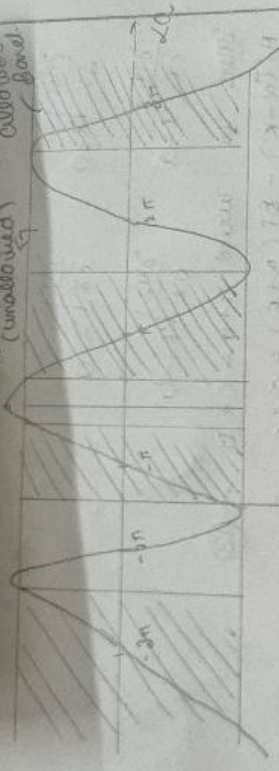


Here the values of potential wave and cos ka takes values between +1 and -1 (forbidden band (unallowed))



To find energy of particle consider $p = \alpha$ then eqn (18) becomes $\sin \alpha a = 0$ and we know $\sin n\pi = 0$.

$$\alpha a = n\pi \Rightarrow \alpha = n\pi/a \Rightarrow d^2 \psi / dx^2 = -\alpha^2 \psi$$

from eqn (3), $\alpha^2 = 2mE / \hbar^2$

then $\frac{n^2 \pi^2}{a^2} = \frac{2mE}{\hbar^2}$

$$\frac{n^2 \pi^2 \hbar^2}{a^2 2m} = E \Rightarrow \frac{n^2 \pi^2 \hbar^2}{a^2 2m} = E \quad \therefore E = \frac{n^2 \pi^2 \hbar^2}{a^2 2m}$$

discrete nature of particle state insulating nature of solid consider $p = 0$ then eqn (18) becomes $\cos \alpha a = 0$

we know, $\cos \alpha a = 0 \Rightarrow \alpha^2 = \frac{n^2 \pi^2}{a^2}$

also $d^2 \psi / dx^2 = -\alpha^2 \psi$ (as per de Broglie wave eqn)

$$K^2 = \frac{h^2}{2m} \alpha^2 = \frac{h^2}{2m} \frac{n^2 \pi^2}{a^2}$$

then $\frac{2mE}{\hbar^2} = \frac{4\pi^2 m^2 v^2}{\hbar^2}$ & $\alpha^2 = \frac{2mE}{\hbar^2}$ (from eqn)

$$E = \frac{4\pi^2 m^2 v^2 \hbar^2}{4\pi^2 2m \cdot \hbar^2} \quad (\therefore \hbar = h/2\pi)$$

$$E = \frac{1}{2} m v^2$$

Above indicate kinetic energy of particle and state conducting nature of solid i.e. particle is completely free and no

Energy level exist

$$u_1(x) = 0 = u_2(x) = 0.$$

$$u_1(x) = a = u_2(x) = -b$$

$$\frac{\partial u_1}{\partial x} \Big|_{x=0} = \frac{\partial u_2}{\partial x} \Big|_{x=0}$$

$$\frac{\partial u_1}{\partial x} \Big|_{x=a} = \frac{\partial u_2}{\partial x} \Big|_{x=-b}$$

Then we write $a+b = c+d$ — (12)

$$A i(\alpha - \kappa) - B i(\alpha + \kappa) = C(B - i\kappa) - D(B + i\kappa) \quad (17)$$

$$A e^{i(\alpha - \kappa)a} + B e^{-i(\alpha + \kappa)a} = C e^{-(\beta - i\kappa)b} + D e^{-(\beta + i\kappa)b} \quad (18)$$

$$A i(\alpha - \kappa) e^{i(\alpha - \kappa)a} - B i(\alpha + \kappa) e^{i(\alpha + \kappa)a} = C(B - i\kappa) e^{-(\beta - i\kappa)b} - D(B + i\kappa) e^{-(\beta + i\kappa)b} \quad (19)$$

for eq (12), (19), (15), (16) a non zero solution exist only if the determinant of the coefficient ABCD vanishes

$$0 = \begin{vmatrix} i(\alpha - \kappa) & -i(\alpha + \kappa) & (B - i\kappa) & -(B + i\kappa) \\ e^{i(\alpha - \kappa)a} & e^{-i(\alpha + \kappa)a} & e^{-(\beta - i\kappa)b} & e^{-(\beta + i\kappa)b} \\ i(\alpha - \kappa) e^{i(\alpha - \kappa)a} & -i(\alpha + \kappa) e^{i(\alpha + \kappa)a} & (B - i\kappa) e^{-(\beta - i\kappa)b} & -(B + i\kappa) e^{-(\beta + i\kappa)b} \end{vmatrix}$$

By showing the determinant, we get $\frac{B^2 + a^2}{a^2 b} \sinh b$

$$\sinh a + \cos \beta b \cosh \alpha a = \cos \kappa (a+b) \quad \text{or } \beta$$

Again solving the above equation

$$\frac{1}{a} \sinh a + \cos \alpha a = \cos \kappa a \quad (18)$$

Acc to bone theory, we know

$$\psi = u(x)e^{ikx} \quad (4)$$

Diff eq (5) w.r.t x

$$\frac{\partial \psi}{\partial x} = \frac{\partial}{\partial x} [u(x)e^{ikx}]$$

$$\frac{\partial \psi}{\partial x} = e^{ikx} \frac{\partial u}{\partial x} + u(x) \frac{\partial e^{ikx}}{\partial x}$$

Again diff w.r.t x

$$\Rightarrow \frac{\partial^2 \psi}{\partial x^2} = e^{ikx} \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} \frac{\partial (e^{ikx})}{\partial x} + ik [u(x) e^{ikx}]$$

$$ik e^{ikx} + e^{ikx} \frac{\partial u}{\partial x} \quad (8)$$

Substituting equation (8) in (5)

$$e^{ikx} \frac{\partial^2 u}{\partial x^2} + ik e^{ikx} \frac{\partial u}{\partial x} - k^2 u e^{ikx} + ik e^{ikx} \frac{\partial u}{\partial x} + \alpha^2 e^{ikx} u_k = 0$$

$$\Rightarrow \% \text{ by } e^{ikx} \quad u_k = 0$$

$$\frac{\partial^2 u_k}{\partial x^2} + 2ik \frac{\partial u_k}{\partial x} - k^2 u_k + ik \frac{\partial u_k}{\partial x} + \alpha^2 u_k = 0$$

$$\Rightarrow \frac{\partial^2 u_k}{\partial x^2} + 2ik \frac{\partial u_k}{\partial x} - (\alpha^2 - k^2) u_k = 0 \quad (9)$$

Similarly for eqn (8) in (6)

$$\frac{\partial^2 u_k}{\partial x^2} + 2ik \frac{\partial u_k}{\partial x} - (\beta^2 - k^2) u_k = 0 \quad (10)$$

Solution for eq (9) and (8) is

$$u_1 = A e^{(\alpha - k)x} + B e^{-(\alpha + k)x} \quad (11)$$

$$u_2 = C e^{(\beta - k)x} + D e^{-(\beta + k)x} \quad (12)$$

Apply boundary condition to above equation such that

Samiconductor Assignment

Kronig-Penny model

- It is assumed in quantum free electron theory of metal that free e⁻s in metal express a constant potential and in free to move in the metal
- Bonds on this many phenomena were explained why some bonds could not be explained why some are conductors / semiconductors / insulators.
- hence the failure was overcome by band zone theory which state that electron move in a periodic potential produced by ion core i.e.
 - (V) is maximum when e⁻ is lying between adjacent nuclei
 - (V) is zero near the ion core (oo) nucleus

Apply schrodinger's time independent wave equation for two regions

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E-V)\psi = 0$$

for 0a in diagram $V=0$, then

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E-V)\psi = 0 \quad \text{--- (1)}$$

for -b0 in diagram $V=V_0$, then

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E-V)\psi = 0 \quad \text{--- (2)}$$

$$\text{Put } \alpha^2 = \frac{2m}{\hbar^2} (E)\psi \quad \text{--- (3)}$$

$$\text{and } \beta = \frac{2m}{\hbar^2} (V_0-E)\psi \text{ then } (V_0 > E)$$

$$\frac{d^2\psi}{dx^2} + \alpha^2\psi = 0 \quad \text{--- (5)}$$

$$\frac{d^2\psi}{dx^2} - \beta^2\psi = 0 \quad \text{--- (6)}$$

